Principia GUI Specification

Component Inventory, Layout, and Keyboard Navigation

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Abstract

This document specifies the graphical user interface for Principia, derived from the 61-page implementation spec. It catalogues every GUI component, defines the two-layer panel architecture, specifies the scoped keyboard navigation system, and maps each component to the underlying spec feature it controls. The design language is scientific instrument: dense, flat, monospace, zero decoration. The canvas is the primary element; everything else serves it.

Contents

Part I

Design Principles

0.1 Aesthetic

Scientific instrument, not web application. The references are oscilloscope front panels, Blender's properties editor, NASA technical diagrams, and the existing Principia proof-of-concept. Specific rules:

- Monospace type for all controls and readouts. Display/serif type only for the title.
- Flat controls: no rounded corners, no shadows, no gradients, no hover animations.
- High information density: every pixel earns its space.
- Labels left-aligned, values right-aligned, sliders are thin tracks.
- Text over icons. `Z₀ (B) 0.07` is better than a mystery glyph with a tooltip.
- One decorative element: the title and subtitle. Everything else is functional.
- Colour only for semantic meaning: orange = keyboard focus, accent = active axis, red = locked, grey = frozen/disabled.

0.2 Layout architecture

Full-viewport canvas with two sidebar layers on the right.

```
+--------------------------------------------+---------+---------+
|                                            |         |         |
|                                            | LAYER 2 | LAYER 1 |
|                                            | (on     | (always |
|              CANVAS                        | demand) | visible)|
|              (full height)                 |         |         |
|                                            |         |         |
|                                            |         |         |
+--------------------------------------------+---------+---------+
| STATUS BAR                                                     |
+----------------------------------------------------------------+
```

Layer 1 (always visible). Navigation essentials: slice offset sliders, slice basis, orientation. Fixed width (∼280px). Cannot be hidden. This is the mixing desk.

Layer 2 (on demand). Contextual panels: chart selection, inspector, render settings, quality controls, IC lookup. Slides in from the right edge of the canvas when triggered by keyboard or click. Slides out on dismiss. Only one Layer 2 panel open at a time. Width ∼300px. On narrow viewports, overlays the canvas with semi-transparent backdrop.

Canvas. Full viewport minus Layer 1 width minus Layer 2 width (when open). WebGPU render target. Handles mouse interaction (pan, zoom, hover, click-to-lock). Overlay layers for: hover tooltip, hover streamline (shape-sphere chart), locked-pixel indicator, physics overlay annotations.

Status bar. Single line, full width, bottom edge. Dense monospace readout of current state.

Title block. Top-left corner, overlaid on canvas. **Principia** in display type, subtitle in spaced monospace. Small footprint, semi-transparent background.

0.3 Responsiveness

Viewport	Layer 1	Layer 2
Wide (>1400px)	fixed sidebar	slides in, pushes canvas
Medium (900–1400px)	fixed sidebar	overlays canvas
Narrow (<900px)	collapses to bottom sheet	full-screen modal

Part II

Component Inventory

Every GUI component, organised by location. Each entry lists: the component type, what spec feature it controls, its keyboard scope level, and whether it lives in Layer 1, Layer 2, the canvas, or the status bar.

0.4 Layer 1 — Navigation sidebar

Always visible. Collapsible sections with chevrons. Only one heavy section expanded at a time (opening one collapses others). The sliders section is the default open section.

0.4.1 RENDER section

Component	Type	Spec reference	Notes
URL	button	shareable state	copies current view state to clipboard
JSON	button	export	exports current config as JSON
PNG	button	export	exports canvas as PNG
RESET	button	—	resets all controls to defaults

0.4.2 DISPLAY section

Component	Type	Spec reference	Notes
RENDER MODE	dropdown	§5.4, Part V	Diffusion, Event Classification, Outcome, Word Length, Stretch, Arc Length, Energy Drift, Phase, Custom
RESOLUTION	dropdown	§28 defaults	256, 512, 1024, 2048, 4096

0.4.3 SLICE BASIS section

Component	Type	Spec reference	Notes
Preset buttons	button grid (2×2 + 1)	§5.1–5.5	CONFIG B×A, INNER PP, OUTER PA, CONFIG+PA, CUSTOM
H-AXIS	dropdown	§5.5 mixed-axis	which Z_k maps to horizontal
V-AXIS	dropdown	§5.5 mixed-axis	which Z_k maps to vertical
±MAG	slider	zoom/scale	half-width of slice in latent space

0.4.4 SLICE OFFSET Z_0 (10D) section

This is the primary navigation control. Always visible by default.

Component	Type	Spec reference	Notes
ZERO	button	—	sets all offsets to 0
SMALL RANDOM	button	—	randomises small perturbation
±RANGE	slider	—	controls range of all Z sliders
Z_0 (B)	slider + value	β (Jacobi angle)	frozen if chart locks this
Z_1 (A)	slider + value	α (Jacobi angle)	frozen if chart locks this
Z_2	slider + value	config param	
Z_3	slider + value	config param	
Z_4 (PP.X)	slider + value	$p_{\rho,x}$	
Z_5 (PP.Y)	slider + value	$p_{\rho,y}$	
Z_6 (PA.X)	slider + value	$p_{\lambda,x}$	
Z_7 (PA.Y)	slider + value	$p_{\lambda,y}$	
Z_8 (M_1)	slider + value	mass logit 1	frozen in Burrau charts
Z_9 (M_2)	slider + value	mass logit 2	frozen in Burrau charts

Visual states for sliders.

- **Active axis**: highlighted border (accent colour) for the two sliders that correspond to the current H-AXIS and V-AXIS.
- **Frozen**: greyed out with lock icon. Visible but not adjustable. Caused by chart constraints (e.g. Burrau locks Z_8, Z_9).
- **Focused**: orange outline (keyboard scope indicator).
- **Default**: normal appearance.

0.4.5 ORIENTATION section

Component	Type	Spec reference	Notes
r (rotation)	slider + value	§8 tilt	global slice rotation angle
Tilt 1	slider + value	§8 tilt	tilt into dimension d_1
Tilt 2	slider + value	§8 tilt	tilt into dimension d_2
Tilt target d_1	dropdown	§8 tilt	which latent dim to tilt into
Tilt target d_2	dropdown	§8 tilt	which latent dim to tilt into

0.5 Layer 2 — Contextual panels

Only one open at a time. Each is triggered by a keyboard shortcut or by clicking a section header that lives at the top of Layer 2's slot.

0.5.1 CHART panel

Trigger: `C` key.

Component	Type	Spec reference	Notes
Chart type	button group	§5.1–5.5	Latent Slice, (L_z, E), (L_z, K), Shape Sphere, Mass Simplex, Burrau ν, Mixed Axis
Shape Sphere sub-controls (visible when Shape Sphere selected):			
Projection	toggle	§5.4	Equirectangular / Equal-area
Pole buffer ε	slider	§5.4	exclusion zone near poles
Streamline on hover	toggle	§5.4	enable/disable hover trajectory
Burrau sub-controls (visible when Burrau selected):			
ν	slider + value	§10	$n/m \in (0, 1)$
(m, n) lookup	text input	§10.1	type integers, snaps to ν
Triple (a, b, c)	read-only	§10	computed from ν
(L_z, E) *sub-controls:*			
$K_{\max}$	slider	§5.2	maximum kinetic energy
γ_K	slider	§5.2	power-law exponent
Mixed Axis sub-controls:			
Axis 1 source	dropdown	§5.5	any physical or latent coordinate
Axis 2 source	dropdown	§5.5	any physical or latent coordinate
Axis 1 warp	dropdown	§5.5	linear / log / power
Axis 2 warp	dropdown	§5.5	linear / log / power

Chart switching behaviour. When chart type changes: (1) update which Layer 1 sliders are frozen, (2) update H-AXIS / V-AXIS labels, (3) invalidate tile cache, (4) run chart validation on current Z_0.

0.5.2 QUALITY panel

Trigger: `Q` key.

Component	Type	Spec reference	Notes
Quality tier	button group	§26	Preview / Balanced / Research
Integrator	dropdown	§9	KDK / Yoshida 4th / Yoshida 6th / RK4
Samples/tile axis	slider	§28	N: 8, 16, 32
Max depth	slider	§28	quadtree depth limit
Horizon T	slider	§9	max integration time
Checkpoints M	slider	§22.10	time checkpoints per trajectory
Ensemble E	slider	§11.3	jittered copies per sample
Decode mode	toggle	§28.3	Auto / Force nonlinear / Force linear

Tier presets. Selecting a tier auto-fills the sub-controls with defaults from the spec's quality tier table. Individual overrides are preserved until the tier button is pressed again.

0.5.3 INSPECTOR panel

Trigger: `Tab` key (only when a pixel is locked).

Component	Type	Spec reference	Notes
IC Summary:			
Masses	read-only 3×value	§15	m_0, m_1, m_2
Positions	read-only 3×2 values	§15	$\mathbf{r}_0, \mathbf{r}_1, \mathbf{r}_2$
Momenta	read-only 3×2 values	§15	$\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2$
Chart coords	read-only	§15	(u, v) in current chart
Latent $\mathbf{z}$	read-only 10×value	§15	full 10D latent vector
Diagnostics:			
Outcome	read-only + colour	§15	class + event code
Free-group word	read-only string	§15, §11.5	e.g. `abBaab` ($\ell = 6$)
t_{end}	read-only	§15	termination time
$d_{\min}$	read-only	§15	minimum pair separation
ϵ_E	read-only	§15	energy drift
L_n	read-only	§15	arc length on S^2
D / D_{rel}	read-only	§15	frequency diffusion
λ_T	read-only	§15	FTLE (if available)
Visualisation panels (embedded canvases):			
Shape sphere	mini canvas	§15.3	animated $\mathbf{n}(t)$ trajectory on S^2
Real space	mini canvas	§15.3	animated $\mathbf{r}_i(t)$ with trails
Controls:			
Precision	toggle	§15.2	f32 (GPU) / f64 (CPU) recompute
Play/pause	button	§15.3	animation control
Speed	slider	§15.3	animation playback speed

0.5.4 IC LOOKUP panel

Trigger: `L` key.

Component	Type	Spec reference	Notes
Input mode	button group	§7	Physical / Latent / Burrau (m, n) / Triple
Physical mode:			
m_0, m_1, m_2	text inputs	§7.1	masses (auto-normalise)
$\mathbf{r}_0, \mathbf{r}_1, \mathbf{r}_2$	text inputs ×6	§7.1	positions
$\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2$	text inputs ×6	§7.1	momenta
Latent mode:			
$Z_0 \dots Z_9$	text inputs ×10	§7.1	raw latent vector
Burrau mode:			
m	text input (int)	§10.1	Euclid parameter
n	text input (int)	§10.1	Euclid parameter
Validation status	read-only	§7.6	shows pass/clamp/reject
LOCK	button	§6	commits lookup and locks

0.5.5 RENDER SETTINGS panel

Trigger: `R` key.

Component	Type	Spec reference	Notes
Render mode	button group	Part V	full list of 30+ modes
Colour palette	dropdown	§21–24	vMF OKLAB, Okabe-Ito, etc.
CVD simulation	dropdown	§21.3	None / Protanopia / Deuteranopia / Tritanopia
κ (vMF concentration)	slider	§21	sharpness of colour blending
Physics overlay	toggle	§24	show/hide BC points, Euler, Lagrange
Stability hue	toggle	§23	stability×hue formula
Colour bar	read-only	—	legend for current render mode

0.6 Canvas overlays

These are drawn on top of the WebGPU canvas, not in the sidebar.

Component	Type	Trigger	Notes
Hover tooltip	floating box	mouse move	IC coords, outcome, word (compact)
Hover streamline	SVG/canvas path	mouse move	shape-sphere chart only
Lock indicator	icon at pixel	click-to-lock	small crosshair or pin
Physics overlay	fixed annotations	toggle	collision points, Euler, Lagrange
Axis labels	text	always	Z_i labels on horizontal and vertical edges
Axis ticks	tick marks + values	always	numerical scale on both axes
Title block	display type	always	Principia + subtitle, top-left

0.7 Status bar

Single line, full width, bottom edge. All monospace, space-separated fields.

```
MODE Diffusion  CHART LatentSlice  ZOOM 1.162  PAN (0.18,0.18)
DEPTH 12/20  TILES 847  SAMPLES 216k  FPS 58  INTEGRATOR Yoshida4
Q1 +1.00·z1 +0.00·z0  Q2 -0.64·z8 -0.63·z9
```

Updates every frame. Shows: render mode, chart type, zoom level, pan position, current/max quadtree depth, tile count, total samples computed, framerate, integrator, and the current slice basis vectors.

When locked, prepend: `LOCKED (0.227, 0.177)`.

Part III

Keyboard Navigation

0.8 Scope system

Keyboard navigation uses a hierarchical scope model. At any time, exactly one scope is active. The orange outline indicates the current scope.

0.8.1 Scope tree

```
ROOT
|---- CANVAS scope
|     |---- arrow keys: pan
|     |---- scroll / +/-: zoom
|     |---- shift+scroll: tilt
|     |---- click: lock pixel
|     |---- hover: tooltip + streamline
|
|---- LAYER 1 scope
|     |---- RENDER section
|     |     |---- [URL] [JSON] [PNG] [RESET] buttons
|     |---- DISPLAY section
|     |     |---- RENDER MODE dropdown
|     |     |---- RESOLUTION dropdown
|     |---- SLICE BASIS section
|     |     |---- preset buttons (2x2+1 grid)
|     |     |---- H-AXIS dropdown
|     |     |---- V-AXIS dropdown
|     |     |---- ±MAG slider → value scope
|     |---- SLICE OFFSET section
|     |     |---- ZERO / SMALL RANDOM buttons
|     |     |---- ±RANGE slider → value scope
|     |     |---- Z0..Z9 sliders → value scope each
|     |---- ORIENTATION section
|         |---- r slider → value scope
|         |---- Tilt 1 slider → value scope
|         |---- Tilt 2 slider → value scope
|         |---- target dropdowns
|
|---- LAYER 2 scope (when open)
    |---- CHART panel sections
    |---- QUALITY panel sections
    |---- INSPECTOR panel sections
    |---- LOOKUP panel sections
    |---- RENDER SETTINGS panel sections
```

0.8.2 Navigation keys

Key	Action	Context
↑ / ↓	move between items at current scope level	all scopes
← / →	adjust value (DAS/ARR)	value scope (sliders)
Enter	drill into focused item	section → control, control → value
Esc	pop up one scope level, or dismiss Layer 2	all scopes
Tab	switch focus: Canvas → Layer 1 → Layer 2	root level

0.8.3 DAS and ARR

Parameter	Default
DAS (Delayed Auto Shift)	200ms
ARR (Auto Repeat Rate)	33ms (∼30 repeats/sec)
Step size (normal)	0.01 per repeat
Step size (shift held)	0.001 per repeat (fine)
Step size (ctrl held)	0.1 per repeat (coarse)

Both DAS and ARR should be user-configurable in settings.

0.9 Global hotkeys

Active regardless of current scope.

Key	Action	Spec reference
`C`	toggle CHART panel (Layer 2)	§5
`Q`	toggle QUALITY panel (Layer 2)	§26
`R`	toggle RENDER SETTINGS panel (Layer 2)	Part V
`L`	toggle IC LOOKUP panel (Layer 2)	§7
`Tab`	toggle INSPECTOR (Layer 2, only when locked)	§15
`M`	cycle render mode forward	Part V
`Shift+M`	cycle render mode backward	Part V
`F`	toggle fullscreen (hide all panels)	—
`P`	export PNG	—
`Space`	pause/resume progressive refinement	—
`/`	open IC LOOKUP with cursor in text field	§7
`.`	cycle quality tier	§26
`1`–`0`	select Z_0–Z_9 for quick adjustment	—
`[` / `]`	nudge selected Z_k down/up	—
`Esc`	unlock pixel / close Layer 2 / pop scope	—

The `1`–`0` plus `[/]` combination enables keyboard-only 10D navigation without entering the slider scope: press `3` to select Z_3, tap `]` repeatedly to sweep that dimension. DAS/ARR applies to `[/]` when held.

Part IV

Interaction Flows

Concrete sequences for common tasks.

0.10 Explore a Burrau configuration

1. Press `C` to open Chart panel.
2. Select **Burrau** ν.
3. Type $m = 2$, $n = 1$ in the lookup fields (snaps to 3–4–5).
4. Press Esc to close Chart panel.
5. Canvas shows the Burrau basin map at rest start.
6. Use `[/]` with `1`–`0` to sweep latent dimensions.
7. Use scroll to zoom into a basin boundary.

0.11 Lock and inspect a trajectory

1. Click on an interesting pixel. Lock indicator appears.
2. Status bar shows `LOCKED (u, v)`.
3. Press `Tab`. Inspector panel (Layer 2) slides in.
4. Shape-sphere mini-canvas shows animated $\mathbf{n}(t)$.
5. Real-space mini-canvas shows animated $\mathbf{r}_i(t)$.
6. Free-group word displayed as string.
7. Toggle f64 precision to validate against f32.
8. Press Esc to close inspector. Pixel remains locked.
9. Press Esc again to unlock. Canvas returns to free mode.

0.12 Shape-sphere exploration with hover streamlines

1. Press `C`, select **Shape Sphere**.
2. Enable **Streamline on hover** toggle.
3. Close Chart panel.
4. Move mouse across canvas. Trajectory streams from cursor.
5. Watch streamline topology change at basin boundaries.
6. Click to lock. Streamline freezes.
7. Press `Tab` for full inspector.

0.13 Mixed-axis phase-space plot

1. Press `C`, select **Mixed Axis**.
2. Set Axis 1: θ (shape). Set Axis 2: L_z.
3. Close Chart panel.
4. Canvas shows shape vs angular momentum basin map.
5. Layer 1 sliders for frozen dimensions adjust the cross-section.

0.14 Deep zoom with keyboard

1. Click to lock a basin boundary.
2. Scroll to zoom. Progressive refinement fills tiles.

3. At depth >20, linearised decoder engages automatically.
4. Status bar shows `DEPTH 25/40 DECODE LINEAR`.
5. Press `Space` to pause refinement at current state.
6. Press `P` to export PNG.
7. Press `Tab` to inspect the locked pixel at f64.

0.15 10D navigation with number keys

1. Press `5` to select Z_5 (PP.Y).
2. Status bar briefly flashes `SELECTED` $\mathtt{Z_5}$ `(PP.Y)`.
3. Hold `]` — DAS delay, then ARR sweeps Z_5 upward.
4. Watch basin structure deform in real time.
5. Press `8` to switch to Z_8 (M_1).
6. Hold `[` — sweep mass ratio downward.
7. Release. Value holds.

Part V

Component Specifications

0.16 Slider

The primary control. Used for all continuous parameters.

Visual structure.

```
Z0 (B)
|----[====|=========]----| 0.07
min                      max  value
```

Elements.

- Label: top-left, monospace, includes physical name in parentheses.
- Track: thin horizontal line, full width of panel minus value box.
- Thumb: small square handle on track.
- Fill: optional filled region from centre to thumb (useful for bipolar sliders centred at 0).
- Value box: right-aligned, monospace, shows current numerical value. Optionally editable on double-click or Enter.
- Range labels: min and max at track endpoints, smaller/dimmer type.

Interaction.

- Mouse: drag thumb, or click track to jump.
- Keyboard (value scope): ←/→ adjust with DAS/ARR. Shift for fine, Ctrl for coarse. Enter to type value directly.
- Scroll (when hovered): adjust value by scroll delta.

States. Normal, focused (orange outline), frozen (greyed, lock icon), active-axis (accent border).

0.17 Dropdown

For discrete selections (render mode, resolution, axis source).

Visual structure.

```
RENDER MODE
[DIFFUSION         v]
```

Interaction.

- Click or Enter: opens option list.
- ↑/↓: navigate options.
- Enter: select and close.
- Esc: close without selecting.
- Typing: filter options by prefix.

0.18 Button and button group

For discrete actions (ZERO, RESET) and mode selection (chart type, quality tier).

Button group renders as a grid of toggle buttons. Exactly one active at a time (radio behaviour). Active button has inverted colours (white text on dark background). Keyboard: ←/→ or ↑/↓ to move selection.

Action button triggers immediately on click or Enter. Brief visual flash on activation.

0.19 Toggle

Binary on/off. Renders as a small button that inverts when active.

`[PHYSICS OVERLAY x]` or `[PHYSICS OVERLAY  ]`

Keyboard: Enter or Space to toggle.

0.20 Text input

For IC lookup and direct value entry. Monospace, minimal styling, no placeholder text — just an empty box with a blinking cursor.

Validation. On Enter, the value is validated against chart constraints (§7.6). If invalid, the border flashes red and the validation status field shows the specific failure. If valid, the value commits and the canvas updates.

0.21 Mini canvas

Embedded WebGL/Canvas2D element within the Inspector panel. Used for shape-sphere trajectory and real-space trajectory animations. Fixed aspect ratio (1:1 for shape sphere, flexible for real space). Shares the locked IC's trajectory data with the main canvas but renders independently.

0.22 Read-only value

For inspector diagnostics. Label left, value right, monospace.

```
OUTCOME     ESC1
WORD        abBaab (l=6)
t_end       67.3
d_min       2.4e-3
eps_E         1.7e-6
```

No interaction. Updates when locked pixel changes or when inspector recomputes.

Part VI

Implementation Notes

0.23 State management

All GUI state lives in a single reactive store. The canvas reads from the store; controls write to it. Store fields map 1:1 to uniform buffer contents — updating a slider immediately writes to the store, which triggers a uniform buffer upload, which triggers a re-render.

Key store fields:

- `chart_type`: enum
- `z0[10]`: float64 array (CPU-side, cast to f32 for GPU)
- `q1[10], q2[10]`: slice basis vectors
- `render_mode`: enum
- `quality_tier`: enum + per-field overrides
- `locked`: null | {uv, z, physical_ic}
- `zoom, pan`: view transform
- `keyboard_scope`: scope tree cursor
- `selected_z`: 0–9 (for number-key quick adjustment)

0.24 Render mode enumeration

Complete list of render modes the RENDER MODE dropdown should expose, grouped by category:

Classification. Event classification (outcome class), event sub-code, escaping body identity.

Scalar fields. Diffusion (D), relative diffusion (D_{rel}), arc length (L_n), event time (t_{end}), minimum separation ($d_{\min}$), energy drift (ϵ_E), stretch / FTLE (λ_T).

Shape-sphere. Phase (unwrapped $\tilde{\theta}$ at final checkpoint), checkpoint hue (vMF blended), direction cosines.

Free-group word. Word length (ℓ), first symbol, abelianisation (n_a, n_b), word hash (lexicographic).

Tile diagnostics. Coherence score, outcome impurity, word agreement, ensemble spread, dominant outcome.

Quality diagnostics. Energy drift heatmap, suspect-sample fraction, decode mode (nonlinear vs linear).

0.25 What is *not* in the GUI

The following spec features are internal and have no direct GUI control:

- SampleSummary / TileSummary struct layout — internal data model.
- Free-group branch-cut placement — derived from collision point positions, not user-configurable.
- Tile-local coordinate transform — automatic, transparent to user.
- Linearised decoder switchover — automatic at depth threshold, shown in status bar only.
- Cache invalidation matrix — internal to quadtree scheduler.
- Workgroup shape — fixed at `@workgroup_size(8,8,1)`.
- WGSL alignment constraints — shader implementation detail.